

PROFESSIONAL SUMMARY:

- Data Engineer with around 8 years of expertise in designing, implementing, and optimizing data architectures. Proficient in a diverse range of programming languages including Python, Scala, SQL, and Bash scripting.
- Expertise in Apache Hadoop ecosystem components like Spark, Hadoop Distributed File Systems (HDFS), Hive, MapReduce.
- Hands on experience in creating real-time data streaming solutions using Apache Spark core, Spark SQL & Data Frames, Kafka, Spark streaming and Apache Storm.
- Migrated an existing on-premises application to AWS. Used AWS services like EC2 and S3 for data sets processing and storage.
- Have good experience working with Azure BLOB and Data Lake storage and loading data into Azure SQL Synapse analytics (DW).
- Exposure and usage of industry standards. Adept at implementing and managing infrastructure as code using Terraform and Cloud Formation.
- Experienced in building and deploying Generative AI models, including LLMs (Large Language Models), text-to-image models, and fine-tuned transformers for domain-specific tasks.
- Hands-on experience using Dimensional Data Modeling, Star Schema Modeling, Snow-Flake Modeling, Fact and Dimension Tables, Physical and Logical Data Modeling.
- Design Develop and test ETL Processes in AWS Glue to migrate Campaign data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift.
- Strong working experience with SQL and NoSQL databases, data modeling and data pipelines. Involved in end-to-end development and automation of ETL pipelines using SQL and Python.
- Worked on migration of data from On-prem SQL server to Cloud databases (Azure Synapse Analytics (DW) & Azure SQL DB).
- Extensive experience in real-time data streaming and processing using technologies such as Apache Kafka, AWS Kinesis, and Azure Event Hubs, enabling low-latency insights for business-critical operations.
- Versatile in working with databases including MySQL, PostgreSQL, MongoDB, Redis, Oracle SQL, CosmosDB, and HBase, ensuring optimized data storage and retrieval.
- Knowledgeable in cloud data warehousing technologies such as Snowflake, Redshift, and BigQuery, enabling high-performance data analytics.
- Acquainted with data serialization formats like JSON, XML, and XSLT for efficient data exchange and transformation.
- Hands-on experience with containerization (Docker) and orchestration (Kubernetes) for scalable deployment of data applications.
- In depth understanding/knowledge of Hadoop Architecture and various components such as HDFS and MapReduce Programming Paradigm, High Availability and YARN architecture.
- Proficient in leveraging OpenAI, Hugging Face Transformers, LangChain, and other GenAI frameworks to design intelligent and scalable solutions.
- Extensive working knowledge in setting up and running Clusters, monitoring, Data analytics, Sentiment analysis, Predictive analysis, Data presentation with big data world.
- Excellent interpersonal and communication skills, creative, research-minded, technically competent, and result-oriented with problem solving and leadership skills.
- Demonstrated ability to integrate GenAI capabilities into real-world applications such as chatbots, content generation tools, summarization engines, and code assistants.
- Hands on experience in setting up workflow using Apache Airflow and Oozie workflow engine for managing and scheduling Hadoop jobs.
- Automated resulting scripts and workflow using Apache Airflow and shell scripting to ensure daily execution in production.

TECHNICAL SKILLS:

Programming Languages	Python, Scala, SQL
Cloud Services	AWS (S3, Glue, Redshift), Azure (ADF, Synapse)
Big Data Technologies	Hadoop, Spark, Databricks, Snowflake
Databases	PostgreSQL, Oracle, MySQL, MongoDB
DevOps Tools	Jenkins, Docker, Kubernetes
ML/GenAI Tools	Scikit-learn, OpenAI, Hugging Face
Orchestration	Airflow, Step Functions, Oozie
On-prem Hadoop	Cloudera, HortonWorks
ETL and Data Modeling	Glue, Informatica, Star Schema
File Formats	Parquet, Avro, JSON, XML
Visualization Visualization	Tableau, Looker, Power BI
Documentation	Confluence, SharePoint
Version Control	GitHub, Bitbucket
SDLC	Agile, JIRA

SOFT SKILLS:

Team player, Quick learner, Communication.

PROFESSIONAL EXPERIENCE:

Bayer, St.louis, MO
Senior Data Engineer

September 2023 to Present

Responsibilities:

- Spearheaded end-to-end implementation of SDLC methodologies, ensuring seamless development, testing, and deployment of data processing applications.
- Extracted data from CSV, JSON and parquet files from S3 buckets in Python and load into AWS S3, Snowflake DB and Snowflake.
- Designed and implemented large-scale data processing pipelines using Hortonworks Data Platform (HDP), leveraging HDFS, Hive, Pig, and Spark for ETL workflows.
- Worked on implementing Lake House architecture from Databricks and used features such as Unity Catalog, DLT, DB-SQL and Delta tables.
- Woked on optimizing spark jobs for saving the compute costs and reducing the overall run time of the pipelines.
- Built and deployed Retrieval-Augmented Generation (RAG) pipelines using LangChain, FAISS, and OpenAI APIs for knowledge-based chatbots.
- Developed scalable ETL pipelines using Apache Spark and Hive on Cloudera Data Platform (CDP), enabling real-time analytics across multiple business units.
- Evaluated existing OLAP cubes and understood dependent fact tables, dimension tables and derived views. Created star schema data model for the same.
- Integrated structured and unstructured data from multiple sources into Hortonworks Hadoop ecosystem using Sqoop, Flume, and Kafka.
- Used Pandas and NumPy in Python scripts for data transformation during S3 to Redshift workflows using AWS Glue.
- Integrated Apache Kafka with Cloudera for event-driven architecture, supporting near real-time data processing and monitoring.
- Experience on writing python scripts to handle metadata throughout pipeline and create Glue Catalog using AWS Lambda.
- Integrated GenAI capabilities into customer-facing applications, enabling dynamic content generation and personalization at scale.
- Experience on writing Redshift queries to create data products and performance tuning and optimization like distribution keys, Redshift Spectrum
- Design and implement scalable data models and schemas within Snowflake to support analytical and reporting requirements.
- Managed vendor relationships and API integrations (OpenAI, Anthropic, Cohere) to deliver low-latency GenAI solutions.
- Created and managed Glue Data Catalogs and Crawlers to catalog metadata from diverse data sources, enabling efficient data discovery and queryability via Athena.
- Optimize Snowflake queries for performance and cost efficiency using best practices such as clustering, partitioning, and query profiling.
- Integrated AWS Glue with other AWS services (S3, Lambda, Redshift, and CloudWatch) to build robust, event-driven data processing solutions with real-time monitoring and alerting.
- Installed, configured, and maintained multi-node Hortonworks Hadoop clusters in both on-premises and cloud environments.
- Architected end-to-end big data platforms using Cloudera CDP, integrating streaming, batch, and ML workloads into a unified platform.
- Collaborated with cross-functional teams to implement AI-assisted features using Hugging Face Transformers, OpenAI SDK, and VectorDBs.
- Developed infrastructure as code (IaC) using Terraform and AWS CDK to manage and provision multi-region environments across dev, staging, and production.
- Leverage Snowflake features such as Time Travel, Zero-Copy Cloning, and Data Sharing to improve data integrity, versioning, and collaboration across teams.
- Automated blue/green deployments for ECS services using AWS CodeDeploy and Lambda, ensuring zero-downtime releases.
- Implemented role-based access control (RBAC) and Kerberos authentication to ensure secure Hadoop operations on Hortonworks.
- Designed and implemented content moderation pipelines to filter inappropriate LLM outputs using reinforcement learning and safety filters.
- Integrated API-based data sources into AWS pipelines, leveraging Python SDKs (e.g., Boto3) and handling authentication, pagination, and error management.

Environment: Python, Scala, Apache Beam, Spark, Hadoop, Kafka, MySQL, MongoDB, ETL, Pandas, NumPy, Step Functions, Snowflake, EKS, ECS, Snowball, GitHub, Amazon Quick Sight, Amazon Elasticsearch, Agile, Scrum, JIRA.

Optum - Eden Prairie, Minnesota
Data engineer

July 2022 to Aug 2023

Responsibilities:

- Evaluated existing OLAP cubes and understood dependent fact tables, dimension tables and derived views. Created star schema data model for the same.
- Extracted data from CSV, JSON and parquet files from ADLS in Python and load into Azure Blob, Snowflake DB and Snowflake.
- Created interactive dashboards by connecting BI tools like Tableau or Power BI to Hortonworks Hive data warehouse.
- Built Spark-based applications for data extraction, transformation, and aggregation from multiple systems, storing processed data in Azure Data Lake using Databricks notebooks.
- Integrated external datasets into a centralized data warehouse using Sqoop and managed large-scale data storage within Hadoop Distributed File System (HDFS).
- Built scalable data pipelines using Azure Data Factory, Databricks, and Synapse Analytics to support enterprise reporting.
- Developed reusable Spark ETL libraries deployed on Cloudera, reducing code duplication and improving job maintainability.
- Developed real-time ingestion and processing solutions using Kafka + Storm/Spark Streaming within the Hortonworks ecosystem.
- Tailored and deployed JSON-based scripts for ADF pipelines, automating SQL Activity for efficient data processing.
- Designed and implemented modern data solutions using Azure PaaS services, enabling advanced data visualizations while assessing impacts on current business processes.
- Created data pipelines using Azure Data Factory (ADF) to transfer parquet files from ADLS Gen2 to Azure Synapse Analytics, ensuring seamless data flow.
- Implemented PostgreSQL and Redis database solutions to enhance data storage and retrieval efficiency for highavailability systems.
- Migrated on-premise applications and databases to Azure using Azure Migrate, Azure Site Recovery, and Database Migration Service (DMS), ensuring minimal downtime.
- Customized predictive machine learning models using Scikit-Learn, providing data-driven solutions for key business decisions.
- Developed scalable data lake solutions using Azure Data Lake Storage Gen2, Azure Synapse Analytics, and Databricks for advanced analytics and reporting.
- Deployed and managed Hadoop clusters on Hortonworks Data Platform 3.x, with custom tuning for high availability and performance.
- Implemented role-based access control (RBAC), Managed Identities, and Azure Key Vault for secure authentication and authorization across services.
- Worked with Pandas and Requests libraries to handle REST API-based data ingestion within Azure Data Factory pipelines.
- Leveraged Apache Flink and Kafka on Cloudera for high-throughput stream processing with exactly-once semantics.
- Integrated OAuth-based authentication to secure API integrations and enhance data security protocols across various external platforms.
- Monitored performance and availability of cloud services using Azure Monitor, Log Analytics, and Application Insights.
- Supervised project workflows and documentation with Kanban, Confluence, and JIRA, fostering efficient project collaboration and task management.
- Automated build and dependency management processes using Maven, enhancing development efficiency and reducing deployment errors.

Environment: Azure, Python, Azure Event Hubs, Hadoop,Blob Storage, ADLS Gen2, SQL Database, HDInsight, Data Factory, Azure Data bricks, Azure Synapse analytics, Spark, pyspark, Docker, Kubernetes, Bitbucket, Jenkins, Spark, Hive, JIRA, Service Bus.

Tailored Brands - Houston, Texas
Data Engineer

Feb 2021 to June 2022

Responsibilities:

- Develop a bigdata web application using agile methodology in Scala as Scala has the capability of combining functional and object-oriented programming.
- Integrated security scans into CI/CD using AWS Inspector, GuardDuty, and third-party tools like Snyk and SonarQube, enhancing pipeline compliance.
- Use Spark to process the data before ingesting the data into the HBase. Both Batch and real-time spark jobs were created using Scala.
- Use HBase as the database to store application data, as HBase offers features like high scalability, distributed NoSQL, column oriented and real-time data querying to name a few.
- Deployed AWS Config, Security Hub, and custom Lambda-based remediation scripts for continuous compliance with CIS AWS Foundations Benchmark and GDPR standards.
- Established end-to-end log analysis and monitoring solutions using ELK stack, enabling comprehensive visibility into system behavior and performance.
- Designed and implemented AWS Transit Gateway and VPC peering for inter-region communication, improving throughput and simplifying network management.

- Implemented strategies for scaling ELK stack components horizontally to handle growing data volumes and ensure consistent performance.
- Managing multi-cloud and hybrid cloud environments using Terraform for infrastructure provisioning and management.
- Integrated AWS environment with on-prem data centers using Direct Connect and IPsec VPNs, ensuring secure hybrid cloud connectivity.
- Implemented a predictive cost modeling solution using AWS Cost Explorer, Athena, and QuickSight to forecast cloud spend and allocate budgets across departments.
- Create and update Jenkins jobs to develop pipelines to deploy the application in different environments like develop, QA and Production.
- Orchestrated chaos engineering exercises using AWS Fault Injection Simulator to validate the resilience of microservices under failure scenarios.
- Participate in requirement gathering and analysis phase of the project in documenting the business requirements by conducting workshops/meetings with various business users.

Environment: SPARK, Scala, Python, Play Framework, PostgreSQL, AWS CLI, Terraform,, SBT, HBase, Hadoop, Hive QL, Hive, HBase, Tableau

HDFC Bank | India
Big Data Engineer

May 2017 to Oct 2020

Responsibilities:

- Used Sqoop for importing the data into HBase and Hive, exporting result set from Hive to MySQL using Sqoop export tool for further processing.
- Created Kafka consumer to store streaming data to HDFS and Designed and implemented the migration from on Premises to cloud environment
- Implemented Spark application to push parquet files into AWS S3 Buckets. Developed a Snowflake warehouse for the Central Repository
- Created Glue jobs using Jupyter Notebooks and Pyspark to process data into s3Buckets. Used Lambda Functions to trigger Glue jobs when data files were stored into s3 Buckets.
- Create Spark application using AWS EMR to process the data and store it into Hive and S3 Buckets. Used Lambda Functions to trigger a submit to EMR cluster into Hive.
- Involved in various phases of development analyzed and developed the system going through Agile Scrum methodology.
- Implemented a generic ETL framework with high availability for bringing related data for Hadoop & Cassandra from various sources using spark.
- Built self-service deployment tooling with CloudFormation and AWS Service Catalog, reducing dependency on DevOps engineers for infrastructure provisioning.
- Exported the analyzed data to the relational databases using Sqoop for visualization and to generate reports for the BI team Using Tableau.
- Experience in developing Spark applications using Spark-SQL in Databricks for data extraction, transformation, and aggregation from multiple file formats for Analyzing& transforming the data to uncover insights into the customer usage patterns.
- Created automated pipelines Using AWS Step Functions and State Machines to deploy EMR Jobs Using Services like EC2, CloudWatch, CodeBuild ,CodeDeploy , S3
- Develop Spark applications using pyspark and spark SQL for data extraction, transformation, and aggregation from multiple file formats for analyzing and transforming the data uncover insight into the customer usage patterns.
- Created automated pipelines Using AWS Step Functions and State Machines to deploy EMR Jobs Using Services like EC2, CloudWatch, CodeBuild , CodeDeploy , S3
- Developed modular DBT models on Snowflake to support real-time analytics for fraud detection, payments, and customer insights.
- Amazon IAM service enabled to grant permissions and resources to users. Managed roles and permissions of users with the help of AWS IAM Services.
- Develop processes for extracting, cleansing, transforming, integrating, and loading data into data warehouse database.
- Build Python Programming to extract data from AWS S3 and load into SQL server for one of business teams as they are not exposed to cloud

Environment: Hive, Snowflake, Oozie, python, DBT, Spark, Jenkins, Spark SQL, Sqoop, NoSQL, Scala, Cassandra, AWS, EMR, EC2, S3, AWSGlue, Kubernetes, Docker.